Transfusion Policy Manual

Clinical Indications for Transfusion & Single Unit Guidelines

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Clinical staff

Introduction

There are risks associated with transfusing patients with blood and blood products. When considering whether transfusion is appropriate, clinicians need to ensure that the benefits outweigh the risks. Transfusion should not be considered the default treatment and available alternatives should be considered if appropriate to reduce the risk of avoidable patient harm.

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☐	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☒	Staff Safety	☐
Other:	☐		

Policy Principles

SCGOPHCG is committed to ensuring the responsible, sustainable, and appropriate use of blood and blood products. SCGOPHCG Blood Management Committee (BMC) follow the guidelines and advice of the National Blood Authority and Australian Red Cross Lifeblood.

Clinical staff are responsible for ensuring that blood and blood products are prescribed in response to an appropriate indication in accordance with the National Blood Authority Patient Blood Management guidelines.

Clinician checklist

Before prescribing blood products the clinician should consider the following questions:

1. What improvement in the patient's condition am I aiming to achieve?
2. Can I minimise blood loss to reduce the patient's need for transfusion?
3. Are there any other treatments I should give before making the decision to transfuse?

4. What are the specific clinical or laboratory indications for red blood cells for this patient?
5. Does the patient have CHRONIC or ACUTE anaemia?
6. Is the patient symptomatic related to anaemia?
7. Do the benefits of transfusion outweigh the risks for this patient?

In deciding whether to transfuse red blood cells, the patient's haemoglobin level, although important, should not be the sole deciding factor. Patient factors, signs and symptoms of anaemia, ongoing blood loss, the risk of anaemia to the patient and the risk of transfusion should be considered.

Informed consent must be obtained as per the Informed Transfusion Consent Policy where transfusion is indicated.

Indications for Transfusion

The decision to transfuse should be based on the National Blood Authority Patient Blood Management Guidelines.
Red Blood cell transfusion should not be dictated by haemoglobin alone, but based on assessment of the patient's clinical status.
In patients with iron deficiency or depleted iron stores, replacement iron therapy is indicated. Please refer to the pharmacy IV iron and oral iron policies.

NBA Patient Blood Management red cell transfusion evidence-based guidelines in clinically stable patients

Hb >100g/L	Transfusion is likely to be unnecessary and inappropriate. *RBC transfusion in Hb>100g/L is associated with increased mortality in patients with acute coronary syndrome (PBM guide: Medical).*
Hb 70-100g/L	The decision to transfuse patients (with a single unit followed by reassessment) should be **based on the need to relieve clinical signs and symptoms** of anaemia, and the patient's response to previous transfusions. *No evidence to warrant a different approach for patients who are elderly or who have respiratory or cerebrovascular disease*.
Hb<70g/L	RBC transfusion may be associated with reduced mortality and is likely to be appropriate. *Transfusion may not be required in well-compensated patients or where other specific therapy is available*.

Indication for transfusion should be documented on the Blood Product Transfusion Form, as part of a valid prescription, utilising the relevant clinical code from the table below. The indication for transfusion, transfusion plan AND transfusion outcome should be documented in the patients' medical record.

Component	Clinical code	Indication for Transfusion
RBC	1	Active bleeding
	2	Symptomatic anaemia
	3	Bone Marrow Suppression
Platelets	4	Bone Marrow Failure Plt count <$10x10^9$/L in absence of risk factors
	5	Bone Marrow Failure Plt count <$20x10^9$/L in presence of risk factors: fever, sepsis
	6	Plt count < $50x10^9$/L with invasive procedure planned
	7	Surgical/invasive procedure: maintain Plt count >$50x10^9$/L
	8	Platelet dysfunction: medical or drug related
	9	Massive Haemorrhage/Transfusion
FFP	10	Liver Disease in the presence of bleeding or at risk of serious bleeding
	11	Multiple coagulation deficiencies: use specific coagulation factors when available
	12	Plasma exchange procedure
	13	Massive Haemorrhage/Transfusion
	14	[1]Warfarin reversal in clinically significant bleeding: in addition to Prothrombinex
Cryoprecipitate	15	Disseminated Intravascular Coagulopathy (DIC)
	16	Fibrinogen Deficiency
	17	Coagulation factor deficiencies
Other	18	Albumin / IVIg / other plasma derived products

Adapted from the National Blood Authority, Patient Blood Management Guidelines: Module 2 Perioperative and Module 3 Medical. (2012)

Single Unit Guidelines

SINGLE UNIT transfusions should be used for clinically stable patients, who are not actively bleeding. In many cases, a single unit of red blood cells will be sufficient to relieve patients' symptoms of anaemia. A second unit should only be considered after assessing the patient for ongoing clinical signs and symptoms of anaemia. NBA Single Unit Decision Support Tool Appendix 1

Reminders:

- Expected haemoglobin increment rise of approximately 10g/L per unit of red blood cells.
- Iron is required to produce red blood cells- patients with anaemia should have iron studies reviewed.
- Where an iron infusion has been deemed appropriate and treatment provided, iron studies should be checked again to assess patient response no earlier than four weeks post infusion.
- Cause of anaemia should always be investigated and follow up provided as required.

Potential benefits of transfusion:

- Improved oxygen delivery
- Correct haemodynamic instability

Risks of Transfusion:

- Adverse reactions include transfusion associated cardiac overload (TACO) and allergic reactions. Adverse reaction risk information can be located at the Australian Red Cross Lifeblood Adverse Incidents webpage.
- Transfusion transmitted infections (TTI's). All blood donations undergo rigorous testing however some risk remains. Refer to the Australian Red Cross Lifeblood Other Transmitted Infections webpage for further information.

Compliance, monitoring and evaluation

The SCGOPHCG Blood Management Committee (BMC) will be responsible for monitoring and reporting compliance against this policy. As a Key Performance Indicator (KPI) an annual single unit RBC prescribing audit will be performed and compliance reported to the SCGOPHCG BMC. The aim is to achieve a compliance rate of greater than 70% for all stable, non-bleeding patients who have received a red cell transfusion, to ascertain adherence to single unit guidelines.

Related Documents

Authorising Policy and Standard/s:

- NSQHS Standard 7 Blood Management

Standards, procedures, guidelines:

- ANZSBT Guidelines for the Administration of Blood Products
- NBA Patient Blood Management Guidelines
- NBA Single Unit Transfusion Guide

Supporting documents

- Blood Product Transfusion Form

References

1. Australian and New Zealand society of blood transfusion, Australian College of Nursing. Guidelines for administration of blood products (Internet). Sydney: Australian& New Zealand Society of Blood Transfusion Ltd; 2019 Oct (cited 2022 Dec 21). Available from: https://anzsbt.org.au/guidelines/guidelines-for-the-administration-of-blood-products/
2. National Blood Authority Australia. Patient Blood Management Guidelines: Module 3 - Medical (Internet). Canberra: National Blood Authority; 2012 Sept (cited 2023 Mar 16). Available from: https://www.blood.gov.au/module-3-medical-patient-blood-management-guidelines
3. National Blood Authority Australia. Single Unit Transfusion Guide (Internet). Canberra: National Blood Authority. (cited 2023, Mar 30). Available from https://www.blood.gov.au/single-unit-transfusion

Authorisation & Version Control

Policy Sponsor	Medical Co-Director Cancer, Imaging and Clinical Services				
Policy Contact	CNC Transfusion				
Date First Issued:	01/12/2013	Last Reviewed:	10/02/2026	Review Date:	10/02/2029
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Approved by:	SCGOPHCG Blood Management Committee			Date:	10/02/2026
Endorsed by:	SCGOPHCG Blood Management Committee			Date:	10/02/2026
Standards Applicable	NSQHS Standards (Version 2):				
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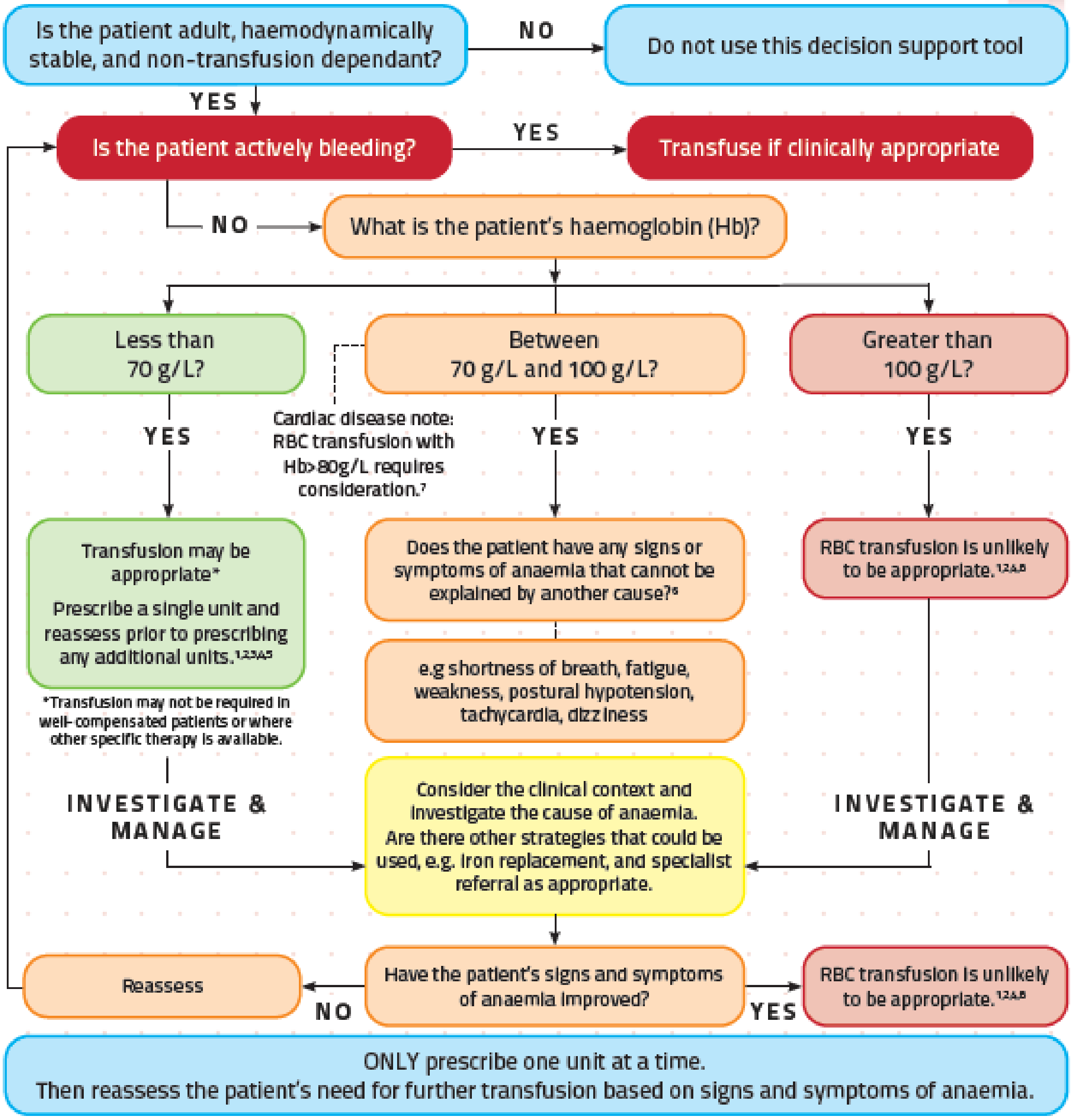
Appendix 1

SINGLE UNIT TRANSFUSION DECISION SUPPORT TOOL

Every UNIT matters

For each red blood cell unit



Rationale— Patient Blood Management Guidelines
Module 2– Perioperative PP2[5], PP3[4]
Module 3– Medical PP2[3], PP3[1], PP5[2], PP6[7]
Module 4– Critical Care PP1[6], PP3[3], PP4[8]

For more information
PBM Guidelines https://www.blood.gov.au/pbm-guidelines
Single Unit Transfusion Guide
https://www.blood.gov.au/single-unit-transfusion

For support
Pathology Laboratory .... SCGH 6383 4018, OPH 6457 8089
Haematologist on call .... via switchboard
Blood Management/Transfusion Nurse ....
.... CNC Transfusion 0478 276 410
Other .... CNC PBM 0478 268 366
....

NATIONAL BLOOD AUTHORITY AUSTRALIA

Abbreviations
PBM — Patient Blood Management
RBC — Red Blood Cells; Hb - haemoglobin
To be read in combination with the Patient Blood Management Guidelines